

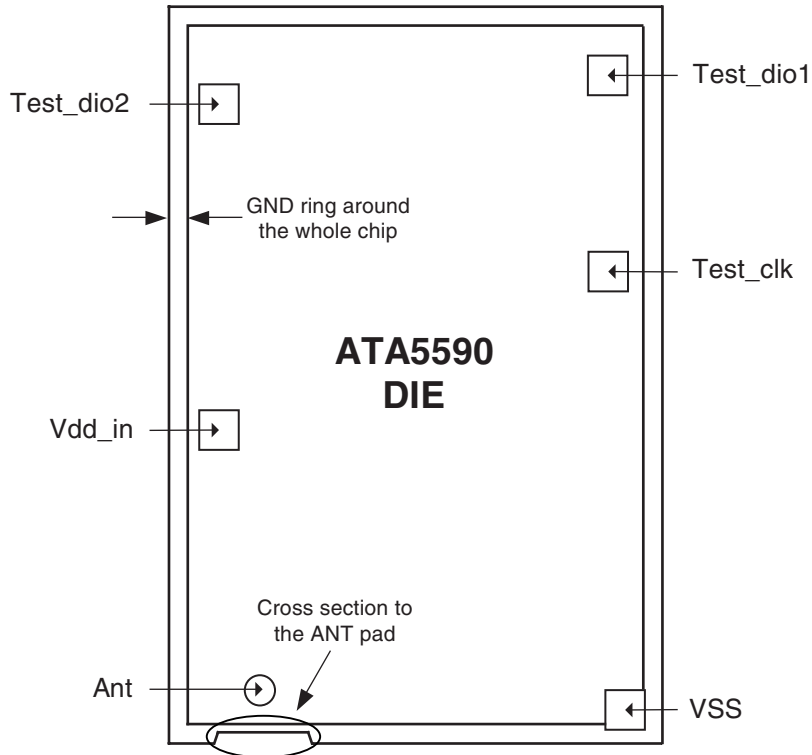


Recommendation for Tag Antenna Connection

1. Overview

The connection to the antenna pad should not cross the GND ring surrounding the whole die too much, in order to avoid coupling capacitances. To reduce the parasitic influence of the antenna connection, the GND ring narrows in the antenna area (antenna pad), as can be seen in [Figure 1-1 on page 1](#), and in [Figure 1-2 on page 3](#) with the exact dimensions.

Figure 1-1. Pad Configuration on the Die (View from Top)



ATA5590 Tag Antenna Connection

Application Note



1.1 Electrical Pad Specifications

ESD:	All pads are protected against ESD damage ANT pad: > 1.5 kV (HBM) Other pads: > 1.5 kV (HBM)
Pad connections:	Test_dio2 and Test_dio1 can be connected together; these two pads can also be considered for backup connection in order to improve the mechanical behavior Note: These test pads should not be connected to the ANT pad Test_clk and Vdd_in should not be connected. Note: These two pads are not appropriate for backup connections, and should also not be connected to the ANT pad

1.2 Mechanical Die Specifications

Die dimensions - including nomenclature line:	1260 μm $\times$ 1890 μm
Die dimensions - excluding nomenclature:	1250 μm $\times$ 1880 μm
Dimensions of the pads on the die:	Diameter of the ANT pad is 60 μm VSS pad is 86 μm $\times$ 114 μm Size of the other pads is 80 μm $\times$ 80 μm
Distance between the pads:	Minimum 250 μm (metal edge to metal edge)
Die thickness:	150 μm with bumps 300 μm without bumps
Pad identification - function:	See Table 1-1 on page 3
Die and pad dimensions:	See Figure 1-2 on page 3
Bump location:	See Figure 1-2 on page 3
Passivation material:	Silicon oxide, thickness: 300 μm Silicon nitride, thickness: 750-nm, UV-resistant layer
Passivation thickness:	1050 nm

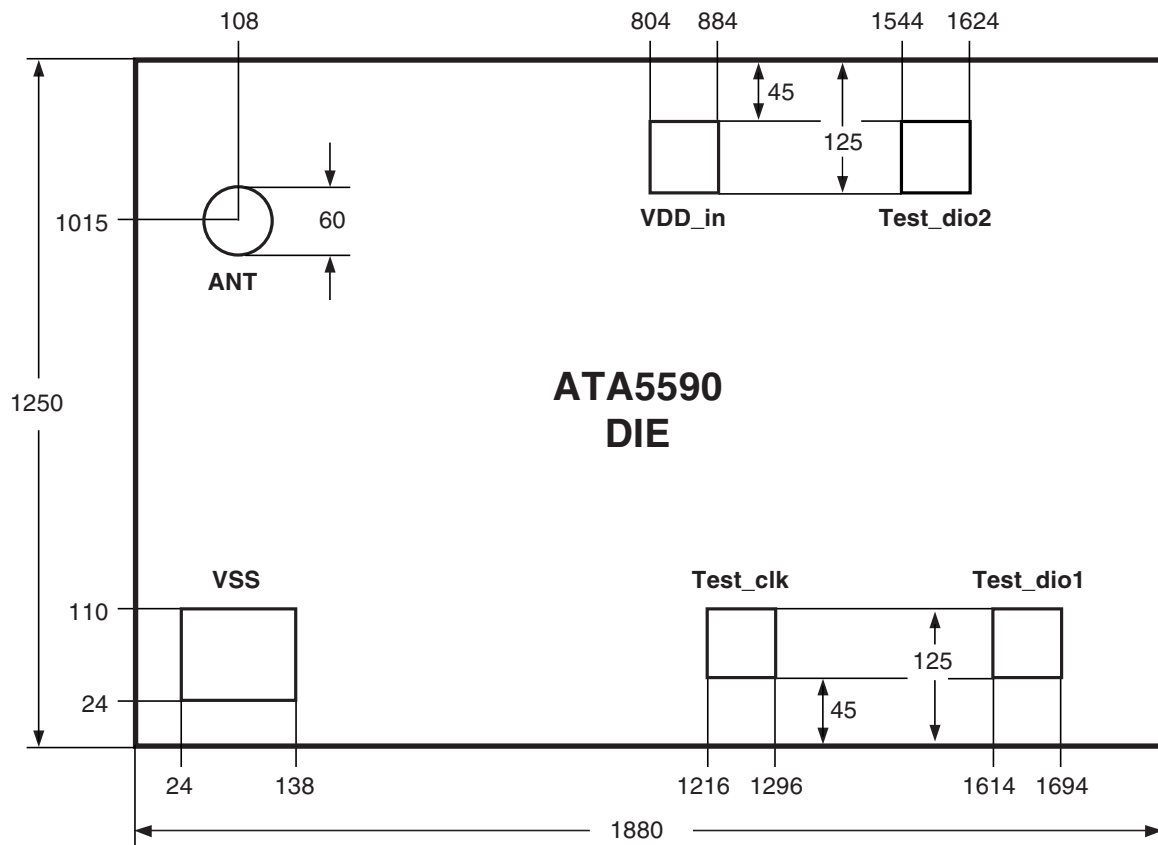
Table 1-1 lists the pads on the die with their functionality.

Table 1-1. Pads on the Default Die

Pad	Function	Physical Construction	Comment
ANT	Antenna	RF in	Diameter of opening 60 μm
VSS	Antenna_gnd	RF ground	Ground pad
VDD_IN	Test pad	Input/output	VDD after the rectifier
Test_Clk	Test clock in	Input	Notch input
Test_dio1	Test-pad	In_out open drain	Control and observation possibilities
Test_dio2	Test-pad	In_out open drain	Control and observation possibilities

Figure 1-2 shows the pad dimensions and location on the die.

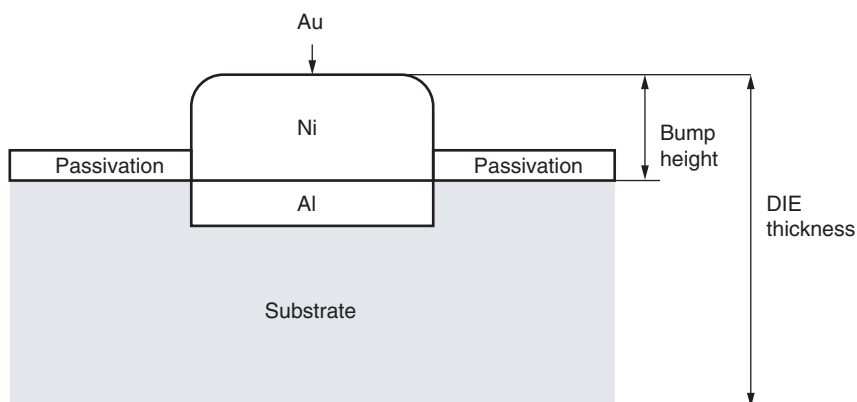
Figure 1-2. Die and Pad Dimensions (in μm)***



1.3 Bump Specifications

Bump material:	Nickel Gold (NiAu) 25- μ m Ni thickness 0.1- μ m Au thickness (coating) See also Figure 1-3
Bump height:	25 μ m (all pads are bumped)
Bump size – Test pads:	80 μ m $\times$ 80 μ m
Bump size – HF pads:	ANT pad: 60 μ m VSS pad: 86 μ m $\times$ 114 μ m
Under bump metallization:	Aluminum (Al)

Figure 1-3. Bump – Lateral Cut



1.4 Mechanical Wafer Specifications

Wafer diameter:	6" (150 mm)
Electrical connection of substrate:	VSS
Charge size:	25

1.5 Wafer Delivery Status

Wafer (die) thickness (ground): 150 μm

300 μm

Failed-die identification:

Each die is electrically tested, failed die are inked

Figure 1-4. Reticle Plan



2. Functional Antenna Connection

In contrast to the VSS pad, the ANT pad significantly influences the read distance of the tag.

Parameters which influence the read distance between reader and tags:

- Tolerance of the connections (ANT pad and VSS pad)
- Coupling capacitances of the cross section of the ANT pad

2.1 Parasitic Capacitance Versus Different Antenna Connections

The antenna connection can be roughly approximated as a plate-type capacitor. Therefore, to calculate the parasitic capacity of the connection, the following formula can be used:

$$C = \epsilon_r \times \epsilon_0 \times \frac{A}{d}$$

2.1.1 Sample Parasitic Capacity Calculation

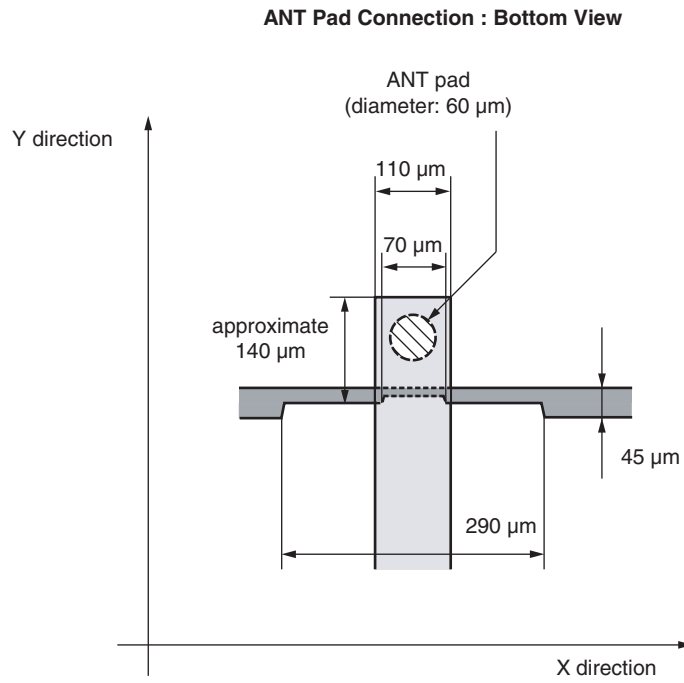
Example: Calculation of the parasitic capacity of the antenna pad connection based on the connection in [Figure 2-3 on page 9](#)

Conductive adhesive ϵ_r :	normally 3 to 4
Dielectric coefficient ϵ_0 :	$8.85 \times 10^{-12} \text{ F.m}^{-1}$
Distance d:	approximately 25 μm (bump height)
ANT feed area:	approximately $140 \times 110 \mu\text{m} = \text{approximately } 15400 \mu\text{m}^2$
ANT pad area:	$60 \mu\text{m}$ (diameter) = $2827 \mu\text{m}^2$

$$A = 15400 - 2827 = \text{approximately } 12573 \mu\text{m}^2$$

$$C = 4 \times 8.85 \times 10^{-12} \times \frac{12573 \mu\text{m}^2}{25 \mu\text{m}} = \text{approximately } 17 \text{ fF}$$

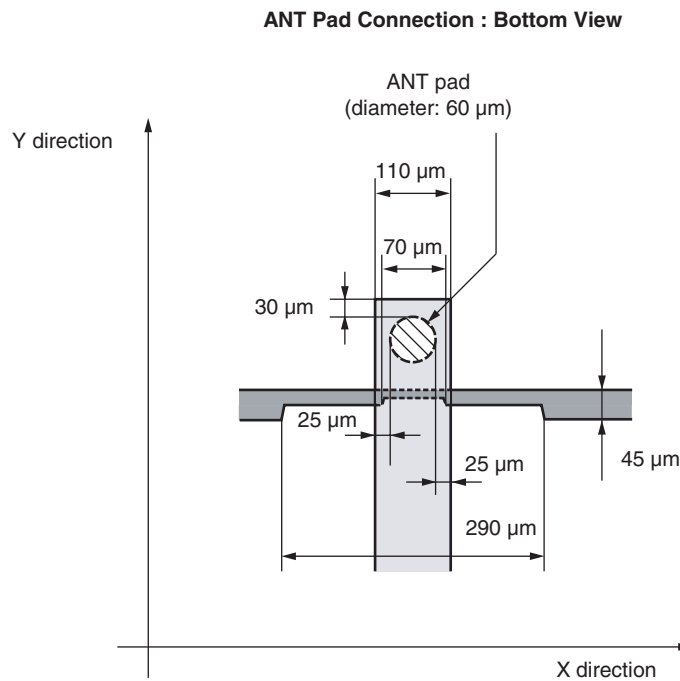
Figure 2-1. ANT Pad Connection Area



2.2 Antenna Connection Recommendation – Assembly Tolerance $\pm 50 \mu\text{m}$ and an Antenna Feeder Width of $110 \mu\text{m}$

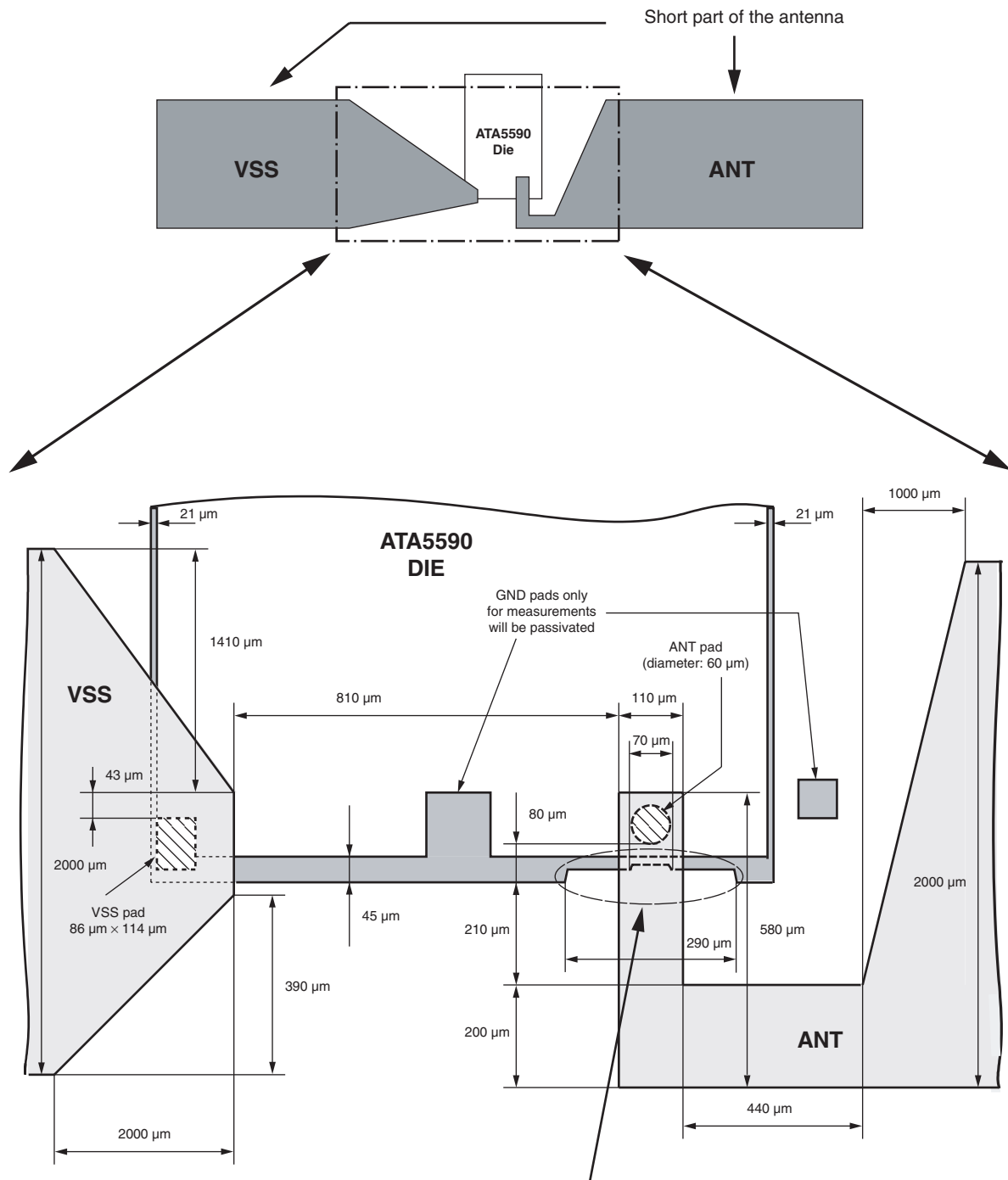
If the assembly tolerance is $\pm 50 \mu\text{m}$, the ANT pad should be connected as shown in [Figure 2-3 on page 9](#). The antenna feeder to the ANT pad should have a width of $110 \mu\text{m}$, thus ensuring a good connection in any case.

Figure 2-2. ANT Pad Connection Recommendation for an Assembly Tolerance $\pm 50 \mu\text{m}$ and an Antenna Feeder Width of $110 \mu\text{m}$



Note: The GND connection can cross the GND ring on a large surface without having any affect on the read distance.

Figure 2-3. Antenna Connection (Bottom View): Assembly Tolerance: $\pm 50 \mu\text{m}$



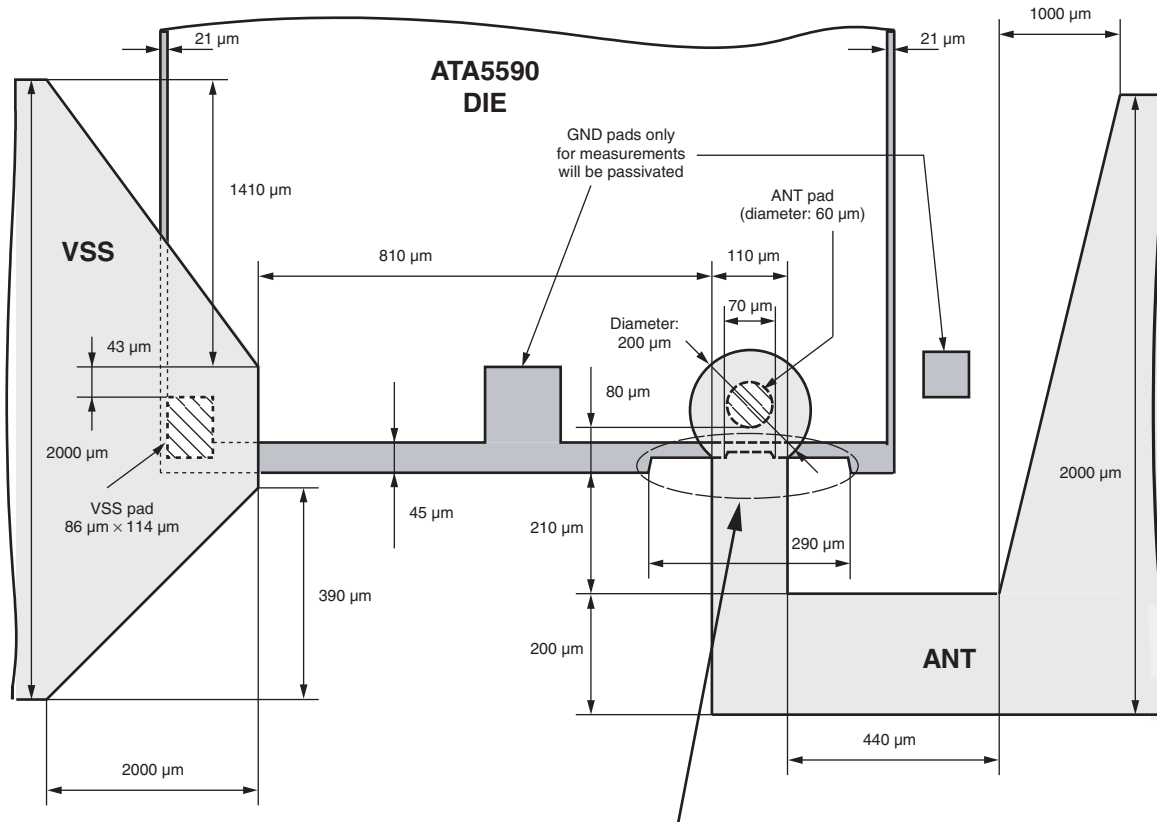
The connection to "ANT" should not cross the chip too much in order to avoid coupling capacitances (there is a GND ring around the whole chip)!!!

Note: Dimensions in this drawing are not to scale.

2.4 Antenna Connection Recommendation – Assembly Tolerance: $\pm 100\ \mu\text{m}$

With this type of connection between ANT pad and antenna, the assembly tolerance can be increased from $\pm 50\ \mu\text{m}$ to $\pm 100\ \mu\text{m}$.

Figure 2-5. Antenna Connection (Bottom View) – Assembly Tolerance: $\pm 100\ \mu\text{m}$



The connection to "ANT" should not cross the chip too much in order to avoid coupling capacitances (there is a GND ring around the whole chip)!!!

- Note:**
1. Dimensions in this drawing are not to scale.
 2. The antenna landing pad area of the ANT pad will be increased significantly, and, therefore, the parasitic capacitance will also be increased, resulting in a shift of the resonance frequency.



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